PROBABILITY 3: ASSIGNMENT 3

Problem 1. For each of the following distributions, find an explicit mapping $T : (0,1) \to \mathbb{R}$ such that $\lambda \circ T^{-1}$ is equal to the given distribution.

(1) Cauchy distribution (density is $\frac{1}{\pi(1+x^2)}$).

(2) Exponential distribution (density is e^{-x} for $x > 0$).

(3) Laplace distribution (density is $e^{-|x|}$ for all $x \in \mathbb{R}$).

(4) A beta density $\frac{1}{\pi\sqrt{x(1-x)}}$ for $x \in (0,1)$.

Problem 2. True or False: Let X, Y, Z be three random variables defined on the same probability space. If $X \stackrel{d}{=} Y$, then $XZ \stackrel{d}{=} YZ$.

(We say that two random variables X and Y have the same distribution, and write $X \stackrel{d}{=} Y$ if the push forward measures of X and Y are the same.)

Problem 3. (Uniform measure on the Cantor set) The Cantor set C is defined by removing $(1/3, 2/3)$ from $[0,1]$ and then removing the middle third of each interval that remains. We define an associated distribution function by setting $F(x) = 0$ for $x \leq 0$, $F(x) = 1$ for $x \geq 1$, $F(x) = 1/2$ for $x \in [1/3, 2/3]$, $F(x) = 1/4$ for $x \in [1/9, 2/9]$, $F(x) = 3/4$ for $x \in [7/9, 8/9], \ldots$

(1) Show that there is a unique continuous extension of F to all of $\mathbb{R}$.

(2) Show that F obtained above is the CDF of a measure μ which does **not** have an atom, and does **not** have a density.

Problem 4. Let X be a random vector with uniform distribution in $\{0,1\}^n$ and let $f, g : \{0,1\}^n \to \mathbb{R}$ be functions such that $f(x)$ and $g(x)$ output the number of 1's and 0's in x respectively. What are the probability distributions of the random variables $f(X) - g(X)$ and $g(X) - f(X)$?

Problem 5. Let U be a random variable whose distribution is Unif$[0,1]$. Consider U^2, does it have a density? If yes, find it.

Problem 6. Suppose that X is a random variable with density f such that $f > 0$ on (a,b) and $f = 0$ outside (a,b) where a, b are such that $-\infty \leq a < b \leq \infty$. Let F_1 be a distribution function (not necessarily that of X). Show that there exists a Borel measurable function $h : \mathbb{R} \to \mathbb{R}$ such that $h(X)$ has CDF F_1.